

# Arpan Pathak

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📺 YouTube: [youtube.com/@ArpanPathak](https://www.youtube.com/@ArpanPathak) | Seattle-based · Local to NVIDIA Redmond (onsite)  
🇺🇸 **Visa**: Requires H1B Transfer sponsorship

## 🎯 SUMMARY

Senior Software Engineer, **8 years**: production-grade **AI inference serving**, distributed **backend microservices**, and cloud-native infrastructure at **Amazon, Microsoft, Oracle**. Strong **Python, C++, Rust, CUDA**; **BERT inference at 5,000 QPS** (ONNX, INT8) on **Docker/Kubernetes** with CI/CD. **Passionate about CV & deep learning for road safety** – edge-AI perception on **NVIDIA Jetson Orin** (TensorRT).

## 🎯 PERSONAL PROJECTS

📁 **Driving-CivicSense: Edge GPU Platform (Rust, CUDA) – NVIDIA Jetson Orin Nano** | [github.com/arpanpathak/driving-civicsense-vision-model](https://github.com/arpanpathak/driving-civicsense-vision-model)

- **YOLOv8/v11 INT8** → **TensorRT** engines at ~ **12 ms** + Deep SORT/Kalman on **NVIDIA Jetson Orin Nano**; real-world 3D scenes, **100%** on-device – quantization and latency tuning on NVIDIA silicon.
- Research paper: **Deterministic Intersection Blockage Prediction** (spatiotemporal) | [paper](#)
- Books: **Seeing Machines · CUDA Kernels** (Rust-to-CUDA) | [arpanpathak.github.io](https://arpanpathak.github.io)

## 🏢 EXPERIENCE

### Software Developer II, Microsoft, Redmond, WA

Jun 2025 – Aug 2026

- **Containerized Cloud Infrastructure**: zero-trust sovereign-cloud networking with eBPF + Cilium; Kubernetes (AKS) operators reconcile state across regions – container-first, policy-driven systems.
- **Systems Performance**: owned the **C++/C#** → **Rust** migration of Microsoft Information Protection; ~ **40% lower P99**, **2x throughput**, **3x lower memory**, six-figure savings.  
↳ **TechStack**: **Rust, C++, Python, Kubernetes (AKS), Docker, CI/CD, Linux**

### Senior Software Engineer, Oracle Cloud Infrastructure, Seattle, WA

Oct 2024 – Mar 2025

- **Backend Services & SDKs**: owned the **Terraform provider** (developer SDK) for Autonomous Database, replacing legacy control-plane tooling; **\$1.2M/month** projected savings.  
↳ **TechStack**: **Java, Go, Rust, REST/gRPC, Terraform, OCI**

### Software Development Engineer II, Amazon, Seattle, WA & Hyderabad, India

Mar 2021 – Oct 2024

- **Production Inference Serving**: scaled low-latency **BERT-based inference** to **5,000 QPS** (ONNX Runtime, INT8 quantization, latency tuning); validated throughput/latency; cut triage SLA **7 days** → **2 hours**.
- **ML Ranking Pipeline**: owned end-to-end (training → serving) XGBoost + Deep Learning ensemble; **5% ad CTR lift**, **\$5.4M/month** revenue.
- **Event-Driven Microservices**: led a team of 3 building ad-delivery microservices at < **50 ms** via real-time CDC (Kotlin).  
↳ **TechStack**: **Python, Kotlin, Java, PyTorch, XGBoost, ONNX Runtime, Kafka, AWS**

### SDE, Razorpay, Bengaluru, India

Aug 2020 – Feb 2021

- **Backend Microservices**: P2P Unified Payments Gateway (Go, Protobuf/gRPC): **3M+ transactions/hour** peak, **99.99% uptime**, **150ms P99**.

## 🔧 SKILLS

↳ **Languages**: Python, C++, Rust, Go, Java, Kotlin, TypeScript, Shell/Bash  
⚡ **AI/ML Inference**: ONNX Runtime, TensorRT (INT8), PyTorch, XGBoost, BERT/LLMs, Quantization, MLOps  
🔧 **Backend & Microservices**: REST, gRPC, Protobuf, Kafka, CDC, High-Throughput APIs  
☁ **Containerization & Cloud**: Docker, Kubernetes (EKS, AKS), Terraform, Kubebuilder, CI/CD, AWS, Azure  
🖨 **GPU & CV**: CUDA, Rust-to-CUDA, GPU Kernels, Jetson Orin, YOLO, 3D Scene Understanding, Edge Inference

## 🎓 EDUCATION

**B.Tech, Computer Science & Engineering**, RCC Institute of Information Technology, Kolkata | 2018